

RAMCO INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Academic Year: 2019-2020(Odd Semester)

INNOVATIVE TEACHING PRACTICES

Degree, Semester& Branch: III Semester B.E. ECE “B Section”

Course Code & Title: EC8393 Fundamentals of Data Structures in C

Name of the Faculty member: Mr. C.A.Yogaraja

Topic: Graph Traversal

Date:13.09.2019

Practice's followed: Think-Pair-Share

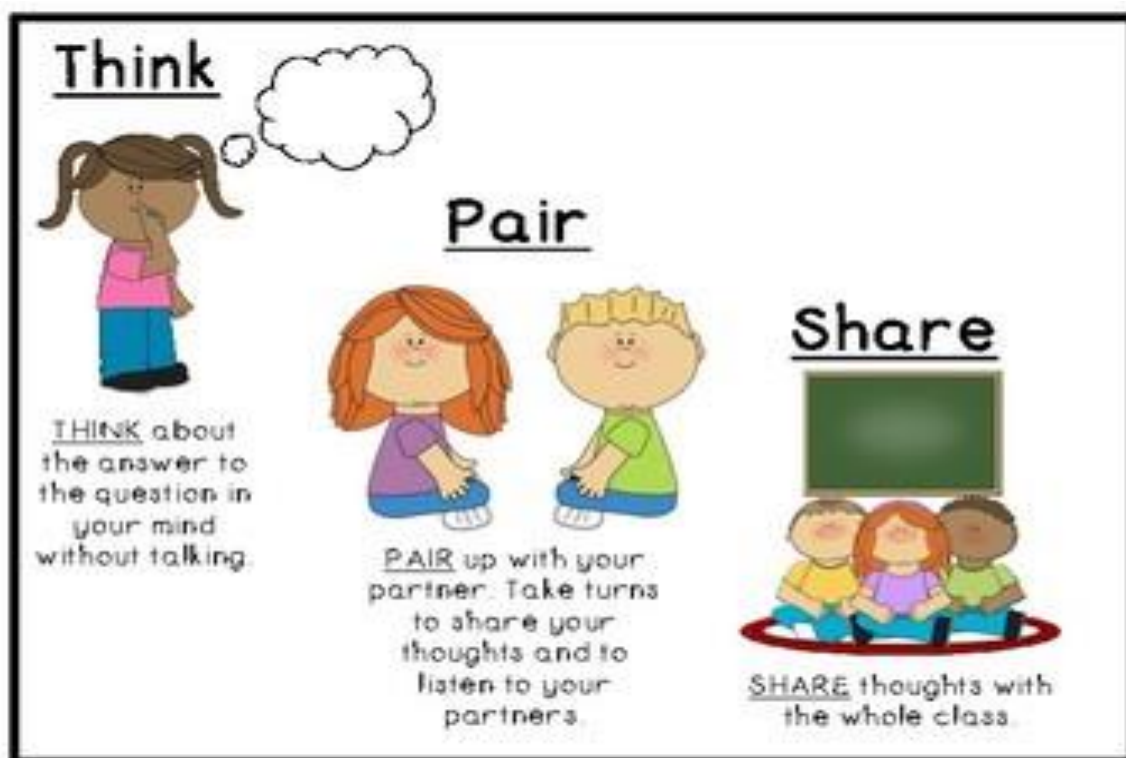
Type of Learning: Active Learning

Learning Objectives:

Objective	Statements
O1	Apply graph traversal algorithms in minimization problems

Think-Pair-Share

Collaborative, active learning strategy, in which students work on a problem posed by instructor, first individually (Think), then in pairs or groups (Pair), and finally together with the entire class (Share).



Use of think-pair-share

- It helps the students to think individually about a topic or answer to a question.
- It teaches the students to share ideas with classmates and builds oral communication skills.
- It helps focus attention and engage students in comprehending the reading material.

Justification for chosen the topic:

Repetition in university question enforces to choose the graph traversal in think Pair Share activity. It will help the students to get out of various misconception and able to find solution for a wide range of problem set.

Implementation of TPS:**Think:**

Students are provided with the various representation of graph(Adjacency matrix and adjacency graph) and they asked to apply the graph traversal algorithm(DFS & BFS) to find the traversal path.

**Pair:**

After the identification of traversal path, the students are allowed to discuss with neighbour and justify the resultant path to each other, if there is any modification to be carried out they allowed to fine tune the path.

Share:

Once the conclusion is made between the pair, then it is to be shared among the whole class by individual one.



OUTCOME:

- Finally, the students are able to gain the knowledge and confidence in finding the graph traversal path, which can be used in minimization problems.

Objective-PO Mapping

Objective	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
O1	3	3	2						3	2	1	1

Reflective Report:

Pre-implementation:

Identified Problems:

- Time management for the activity of presentation from individual team is difficult
- Involvement of students

Initiatives to address the problem:

- Several university questions were shown and discussed to the students about the weightage for graph traversal
- Planned to select the single presentation from individual type of problem.

Post-implementation:

- In a short term of period students had a chance to face variety of problems with solution
- Coordination level among the students increased
- Students came up with variety of doubts while solving the problem

References:

1. Dr.M.Kaliappan “DTD-DTD for TV schedule” in course “IT6801 Service Oriented Architecture” dated: 10.07.2019. [<https://www.ritrjpm.ac.in/departments/computer-science-engg/innovative-practices.php>]
2. Mr.D.Lakshmanan “Use Case Diagram for Real Time Systems” in course “CS8592 Object Oriented Analysis and Design” dated: 04.07.2019. [<https://www.ritrjpm.ac.in/departments/computer-science-engg/innovative-practices.php>]
3. <https://www.teacherspayteachers.com/Product/Think-Pair-Share-2496919>